

Non-invasive continuous lipid profiling via cofactor-refreshing cascading enzymatic reactions and causal machine learning

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José A. Lasalde-Ramírez^{1,6}, Chihyeong Won^{1,2,6}, Sijie Ji^{1,6}, Soyoung Shin¹, Robin M. McDonald¹, Katherine Karish¹, Gwangmook Kim¹, Hong Hang¹, Daniel Mukasa¹, Brian Arianpour³, Yu Song¹, Jieping Yang⁴, Tzung K. Hsiai^{3,4}, Zhaoping Li³, Taeyoon Lee², Juliane R. Sempionatto^{1,5}✉ & Wei Gao¹✉

Continuous lipid profiling could reveal dynamic lipid fluctuations underlying cardiometabolic health, yet existing enzymatic sensing strategies cannot sustain the multi-enzyme reactions required for continuous lipid detection, particularly when non-regenerable cofactors are consumed. Here we introduce a wearable epidermal lipid profiler for non-invasive, continuous monitoring of cholesterol and triglycerides using dual-step and cascading enzymatic reactions supported by the continuous refreshment of adenosine triphosphate (ATP). The system integrates a polymeric cofactor-releasing module composed of polypyrrole–ATP nanostructures that provide controlled, sweat-triggered ATP replenishment for long-term biosensing. We demonstrate real-time lipid tracking during dietary challenges and habitual activities, capturing distinct postprandial responses to different macronutrient compositions. In a preclinical study, simultaneous sweat and blood analyses coupled with causal machine learning revealed key physiological confounders and established individualized mappings between sweat and blood lipids. This cofactor-refreshing platform extends continuous monitoring to complex enzymatic systems, establishing a foundation for non-invasive, longitudinal cardiometabolic health assessment.

Lipid profiling is fundamental to the evaluation and management of metabolic syndrome, diabetes, obesity and cardiovascular diseases¹. Dysregulated lipid levels are key risk factors for atherosclerosis, heart disease and stroke, which remain leading causes of mortality worldwide². In the USA, 48.6% of adults have some form of cardiovascular disease, and the prevalence of obesity reached 41.8% between 2017 and 2020³. As lifestyle factors such as diet, physical activity and smoking strongly influence lipid homeostasis, more accessible and informative

lipid monitoring technologies are needed to support personalized disease prevention and management⁴.

The current clinical standard, periodic blood lipid panels obtained via venipuncture, is invasive, inconvenient and provides only single-point snapshots of systemic lipid levels. These constraints limit patient compliance and obscure dynamic lipid fluctuations. Although alternative microsampling and simplified sensing methods have been explored, they still rely on blood extraction and cannot support

¹Division of Engineering and Applied Science, California Institute of Technology, Pasadena, CA, USA. ²School of Electrical and Electronic Engineering, Yonsei University, Seoul, Republic of Korea. ³Department of Bioengineering, Henry Samueli School of Engineering and Applied Sciences, University of California Los Angeles, Los Angeles, CA, USA. ⁴Department of Medicine, David Geffen School of Medicine, University of California Los Angeles, Los Angeles, CA, USA. ⁵Department of Electrical and Computer Engineering, Rice University, Houston, TX, USA. ⁶These authors contributed equally: José A. Lasalde-Ramírez, Chihyeong Won, Sijie Ji. ✉e-mail: jsemp@rice.edu; weigao@caltech.edu

non-invasive or continuous lipid tracking^{5,6}. To date, no technology has achieved real-time, non-invasive lipid profiling.

Sweat offers a promising alternative to blood as a readily accessible biofluid that reflects nutritional and metabolic status^{7–13}. Recent advances have enabled the continuous tracking of diverse analytes, including metabolites, electrolytes, amino acids and vitamins, through soft, skin-interfaced devices^{13–20}. Among available biorecognition strategies, enzymatic reactions are particularly attractive for continuous sensing because of their specificity, operational stability and catalytic performance. However, continuous lipid monitoring poses unique challenges owing to its reliance on complex cascading enzymatic reactions and non-regenerable cofactors such as adenosine triphosphate (ATP)^{21,22}. Although the presence of cholesterol (CH) and triglycerides (TG) have been detected in sweat²³, existing systems cannot sustain the sequential enzymatic pathways or cofactor availability required for long-term lipid detection.

Translating sweat biomarkers into clinically meaningful insights requires rigorous cross-validation against blood lipids²⁴. Sweat–blood lipid relationships are often confounded by individual physiological and demographic variability and by the complex transport behaviour of hydrophobic lipid species across skin and glandular barriers^{25,26}. Conventional machine learning (ML) approaches that rely primarily on statistical associations fail to capture these causal dependencies^{27,28}. Causal inference frameworks that account for such confounders are therefore critical for establishing reliable, individualized mappings between sweat and blood lipids.

To address these challenges, we developed a wearable epidermal lipid profiler for continuous, non-invasive monitoring of cardiometabolic biomarkers through sweat lipid profiling (Fig. 1a and Supplementary Fig. 1). This platform tracks CH and TG through two distinct enzymatic pathways: a dual-step reaction for CH and a cascading reaction for TG that requires continuous ATP replenishment (Fig. 1b). To sustain cofactor availability, we engineered a polymeric cofactor-releasing (Poly-CORE) unit composed of ATP-encapsulating polypyrrole (PPy)/ATP nanostructures, which provides sweat-activated ATP release within the sensing chamber (Fig. 1c). An integrated multi-walled carbon nanotube (MWCNT)-based ATP sensor enables the voltammetric monitoring of cofactor dynamics, whereas complementary glucose (Glu), pH and sweat rate (SR) sensors provide metabolic context and environmental normalization. A miniaturized iontophoresis (IP) module allows localized sweat secretion on demand (Fig. 1d). The standalone patch is fabricated through scalable processes, including inkjet printing of sensor components and laser engraving of microfluidic layers (Fig. 1e and Supplementary Figs. 2 and 3).

We demonstrate the device's capability to capture dynamic lipid responses to dietary intake and habitual activities. In a 24-participant preclinical study, simultaneous sweat and blood sampling combined with causal ML revealed key physiological confounders and established individualized mappings between sweat and blood lipids (Fig. 1f). This integrated biosensing and modelling framework enhances the accuracy and interpretability of non-invasive lipid profiling for continuous cardiometabolic health assessment.

Biosensor characterization

Three principal amperometric biosensors were developed to quantify Glu, CH and TG, key biomarkers of cardiometabolic health. CH and TG reflect lipid transport, clearance and dysregulation^{29–31}, whereas Glu is closely coupled to lipid metabolism and contributes to dyslipidemia when chronically elevated³². Integrating these enzymatic enables multiplexed profiling of cardiometabolic health.

Glu biosensing was achieved through encapsulating Glu oxidase (GOx) within a bovine serum albumin (BSA)–glutaraldehyde (GT) matrix (Fig. 2a), providing a robust microenvironment for enzyme immobilization³³. The sensing electrode was fabricated by sequentially modifying an inkjet-printed carbon electrode with nanostructured gold

(Au), Prussian blue (PB) and nickel hexacyanoferrate (NiHCF) layers to enhance electroactive surface area³⁴, electrocatalytic efficiency and long-term stability (Fig. 2a). PB served as a redox mediator at low operating potentials around 0 V (refs. 35,36), whereas NiHCF stabilized PB for extended sensing performance³⁷. The same electrode modification and encapsulation protocol was applied across all enzymatic sensors (Supplementary Fig. 4). The Glu sensor exhibited high sensitivity ($66.5 \pm 1.3 \text{ nA cm}^{-2} \mu\text{M}^{-1}$, $n = 5$) and linearity within the physiological sweat concentration range (0–200 μM) (Fig. 2b).

CH detection was enabled through a dual-step enzymatic reaction combining CH esterase (ChEs) and CH oxidase (ChOx) (Fig. 2c). As most epidermal CH exists in esterified form, ChEs hydrolyses cholesteryl esters into free CH and fatty acids, followed by ChOx-mediated oxidation of CH to generate cholest-4-en-3-one and hydrogen peroxide^{7,23}. The sensor demonstrated excellent linearity in the 0–3 μM range, consistent with expected sweat CH concentrations (Fig. 2d and Supplementary Fig. 6) and exhibited a sensitivity of $52.7 \pm 3.7 \text{ nA cm}^{-2} \mu\text{M}^{-1}$ ($n = 5$). This represents a substantial advance, as most reported CH sensors target millimolar concentrations²¹ rather than the low-micromolar regime found in sweat.

TG biosensing relied on a three-enzyme cascade involving lipase (LIP), glycerol kinase (GK) and glycerol-3-phosphate oxidase (G3PO) (Fig. 2e). LIP hydrolyses TG into glycerol and fatty acids; GK phosphorylates glycerol to glycerol-3-phosphate (G3P) in the presence of ATP; and G3PO oxidizes G3P to dihydroxyacetone phosphate and hydrogen peroxide (Supplementary Note 1). As the enzymatic cascade responds to glycerol, endogenous sweat glycerol may contribute to the measured signal. Thus, the sensor does not provide a strictly selective measurement of absolute TG concentration but quantifies glycerol generated from lipid metabolism, including contributions from TG hydrolysis, and therefore serves as a proxy for lipolytic activity. The TG sensor achieved a sensitivity of $100.4 \pm 12.9 \text{ nA cm}^{-2} \mu\text{M}^{-1}$ ($n = 5$) over the 0–150 μM range (Fig. 2f and Supplementary Fig. 6). As ATP is consumed during the reaction, maintaining signal stability requires continuous ATP replenishment, which underscores the need for a controllable cofactor-release module for wearable sensing application.

All sensors demonstrated good reproducibility and shelf stability, maintaining consistent signals over 4 weeks (Fig. 2g and Supplementary Figs. 7 and 8). Greater sensor-to-sensor variability CH and TG arises from the increased sensitivity of multi-enzyme cascades to small variations in fabrication and enzyme loading. The sensors exhibited strong selectivity against common sweat constituents such as uric acid (UA), creatine (CR) and choline (Fig. 2h and Supplementary Fig. 8). Sensor outputs exhibited pH-dependent baseline shifts under physiological variations induced by IP³⁸, with no substantial changes in sensitivity across the tested pH range (Supplementary Fig. 9). The pH effect was more pronounced for the CH sensor because of its lower operating concentration range but remained consistent across devices, enabling systematic correction using an integrated potentiometric pH sensor (Fig. 2i). The influence of physiological temperature variations (Supplementary Fig. 10) and operational stability (Supplementary Fig. 11) were also evaluated. Validation against commercial assay kits confirmed high accuracy for Glu, CH and TG quantification in sweat (Fig. 2j).

To complement biomarker analysis, an impedimetric SR sensor was integrated to assess dynamic sweat generation and analyte dilution. The SR module consists of three electrodes (Fig. 2k): the impedance between E1 and E2 reflects ionic strength, whereas the E2–E3 interdigitated electrodes capture volumetric admittance changes proportional to sweat accumulation and SR. Instantaneous SR is determined from the sweat volume accumulated in the microfluidic channel over time (Fig. 2l and Supplementary Note 2). Sequential introduction of two solutions with differing concentrations produces characteristic impedance shifts in the E1–E2 pair, enabling ionic strength quantification independent of ion identity (Supplementary Figs. 12 and 13). Combining these measurements with volumetric impedance readings from E2–E3 allowed the

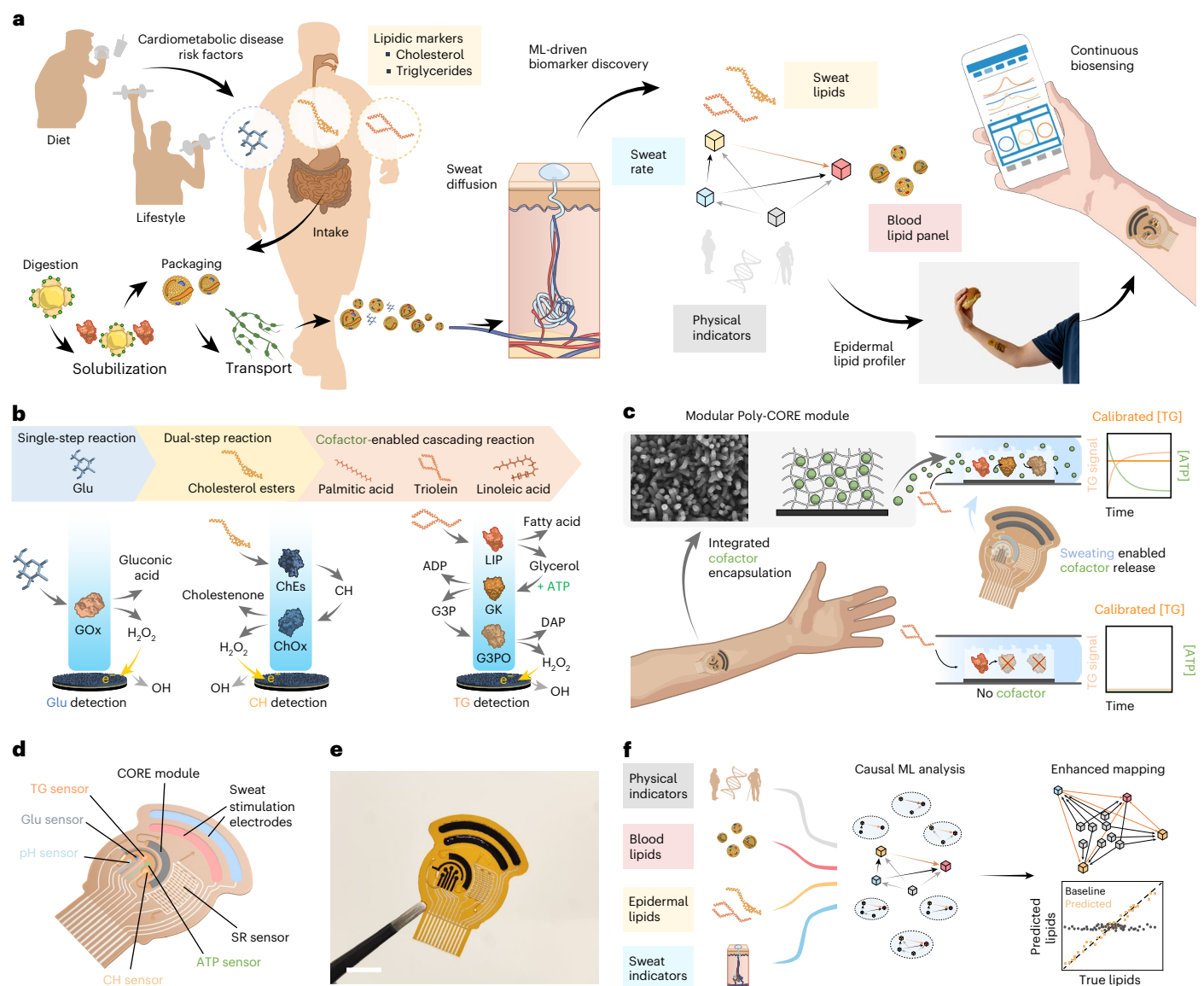


Fig. 1 | Wearable platform for continuous epidermal lipid profiling. a, An overview of key contributors to cardiometabolic diseases—including diet and lifestyle—that modulate systemic lipid metabolism and lead to dysregulation of CH and TG. These lipid biomarkers can be continuously and non-invasively monitored through sweat-based analysis. **b**, A schematic of single-step, dual-step and cofactor-enabled cascading enzymatic reactions leveraged to detect Glu, CH and TG, respectively. **c**, An illustration of the modular Poly-CORE system based on PPy/ATP nanostructures, which enable sustained sweat-triggered ATP

diffusion into the sensing chamber for long-term TG monitoring. **d**, A schematic of the wearable patch design illustrating the bioelectronic components and capabilities. **e**, An optical image of a wearable sensor patch for epidermal lipid monitoring. Scale bar, 10 mm. **f**, The workflow of causal ML-based analysis integrating physical indicators, sweat biomarkers and blood lipid data to identify physiological confounders and enhance individualized mapping between sweat and blood lipid concentrations.

real-time estimation of SR³⁹ (Supplementary Fig. 14). Measurements of sweat impedance as a function of temperature and pH further improved accuracy across physiological conditions (Supplementary Figs. 15 and 16). The SR sensor geometry can be tailored to specific sweat volumes or experiment durations by adjusting the microchannel dimensions (Supplementary Fig. 17). On-body validation using colorimetric flow tracking confirmed the accuracy of the sensor configuration (Supplementary Fig. 18). Together, these capabilities enable continuous SR monitoring for quantitative biomarker normalization.

Sweating-enabled Poly-CORE module

The detection of TG has not previously been demonstrated in a wearable platform because its cascading enzymatic reaction requires a continuous supply of a non-regenerable cofactor. TG biosensing demands

prolonged access to ATP to enable phosphorylation of glycerol to G3P and adenosine diphosphate (ADP) in the secondary enzymatic step²² (Supplementary Fig. 19). To address this challenge, we engineered a Poly-CORE module integrating PPy/ATP nanostructures synthesized via electrodeposition (Fig. 3a). PPy was selected because its redox-active backbone enables electrostatic incorporation of anionic species such as ATP during electropolymerization, forming a porous conductive reservoir that supports sustained cofactor release through ion exchange and diffusion. Under matched fabrication conditions, PPy exhibited a more sustained and gradual ATP release profile after the initial burst phase than other tested polymers (Supplementary Fig. 20). Comparisons with representative polymer entrapment strategies confirmed that dopant-mediated incorporation in PPy provided the most stable release above the biosensing threshold (Supplementary Fig. 21).

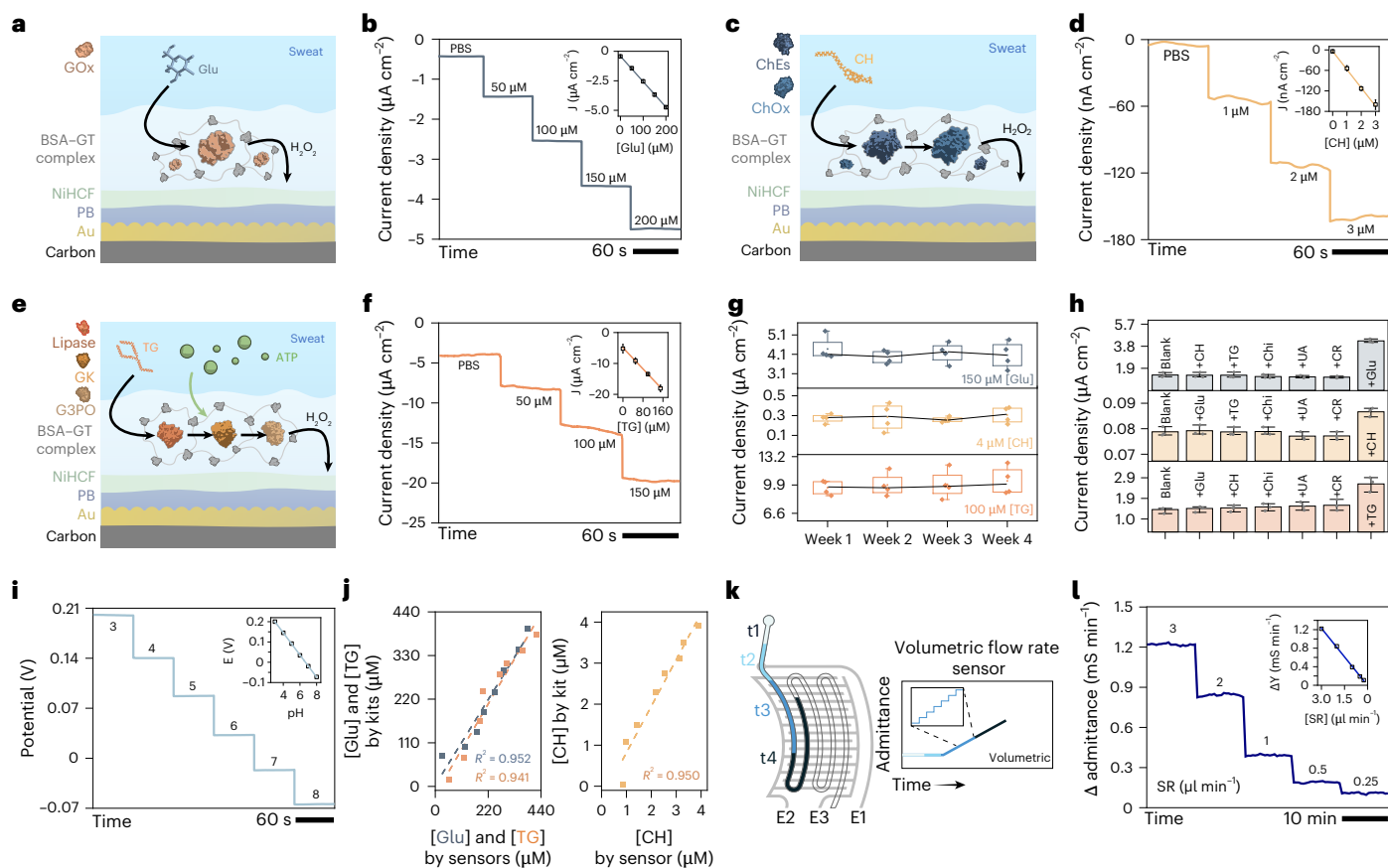


Fig. 2 | Biosensor characterization for continuous lipid monitoring. **a**, The Glu biosensor configuration. **b**, The amperometric response and linear calibration for Glu detection. **c**, The CH biosensor configuration. **d**, The amperometric response and linear calibration for CH detection. **e**, The TG biosensor modification layout. **f**, The amperometric response and linear calibration for TG detection. Error bars in **b**, **d** and **f** represent s.d. from the mean of five sensors. **g**, The 4-week shelf stability of Glu, CH and TG sensors. Error bars represent s.d. from the mean from four sensors. **h**, The selectivity of Glu, CH and TG sensors

against common sweat interferents. [Glu] = 100 μM ; [CH] = 1 μM ; [TG] = 50 μM ; [UA] = 40 μM ; [CHI] = 50 μM ; [CR] = 15 μM . Error bars represent s.d. from the mean from three sensors. **i**, The potentiometric response and calibration curve of the integrated pH sensor. **j**, The validation of Glu, CH and TG sensors against commercial assay kits for analysing sweat samples ($n = 8$). **k**, A schematic of the three-electrode SR sensor. **l**, An admittance variation with flow rate and the corresponding calibration plot. Error bars represent s.d. from the mean from five sensors. Y, admittance.

A thin PPY film was first electrodeposited on an inkjet-printed carbon electrode to promote uniform growth of the PPY/ATP network⁴⁰. By modulating current density and deposition time, distinct PPY/ATP architectures were synthesized with tunable ATP-releasing profiles (Fig. 3b). Electrochemical impedance spectroscopy confirmed increased surface impedance after polymerization, indicating the formation of the cofactor-releasing layer (Supplementary Fig. 22). Scanning electron microscopy (SEM) revealed uniform PPY films (top left) and distinct PPY/ATP morphologies generated by electrodeposition at 0.15 mA cm^{-2} for 1, 10 and 30 min (Fig. 3c).

To optimize the Poly-CORE structures for continuous TG detection, we first determined the limiting ATP concentration for enzymatic activity. The amperometric analysis showed that TG sensors produced distinct signals at physiologically relevant TG levels at ATP concentrations $\geq 100 \mu\text{M}$ (Fig. 3d), a threshold further confirmed under skin-temperature conditions (Supplementary Fig. 23). Thus, the Poly-CORE module was designed to sustain ATP levels above 100 μM .

For continuous operation, the Poly-CORE module was incorporated upstream of the biosensing chamber within the microfluidic platform. As sweat passes through the PPY/ATP nanostructures, ATP diffuses passively into the fluid (Fig. 3e), maintaining cofactor availability. In static conditions, ATP depletion leads to rapid signal decay, whereas dynamic in vitro measurements showed that ATP replenishment restored the TG signal after depletion (Fig. 3f). We next characterized

the long-term ATP release from the Poly-CORE module (Fig. 3g). Optimized PPY/ATP structures sustained ATP concentrations above the biosensing limit for more than 20 h, supporting prolonged operation. The effective operational lifetime depends on SR-driven depletion kinetics. By fixing electrodeposition time to 10 min and varying current density from 0.075 to 1.0 mA cm^{-2} , we generated distinct release profiles (Supplementary Figs. 24 and 25). Modifying electrodeposition solution compositions further tuned cofactor-release kinetics (Supplementary Fig. 25), highlighting the versatility of this approach for other cofactor-dependent enzymatic reactions.

On the basis of the TG sensor's dynamic range, PPY/ATP structures prepared at current densities $\geq 0.15 \text{ mA cm}^{-2}$ provided optimal performance. Flow-dependent studies confirmed that higher sweating rates reduced ATP concentrations in the sensing chamber owing to accelerated mass transfer (Fig. 3h). Nonetheless, the mean ATP concentration remained above the biosensing limit ($1,494 \pm 163 \mu\text{M}$, $646 \pm 111 \mu\text{M}$ and $180 \pm 14 \mu\text{M}$ at flow rates of 0.3, 0.5 and 1.0 $\mu\text{l min}^{-1}$, respectively). We further evaluated ATP release under varying SR cycles and repeated hydration–dehydration cycles, demonstrating consistent release behaviour under conditions mimicking on-body use (Supplementary Figs. 26 and 27). These findings define key design constraints for reliable operation, namely controlled ATP loading and release kinetics that maintain concentrations above the enzymatic threshold across physiological SR variation. As ATP acts as a dopant within the PPY

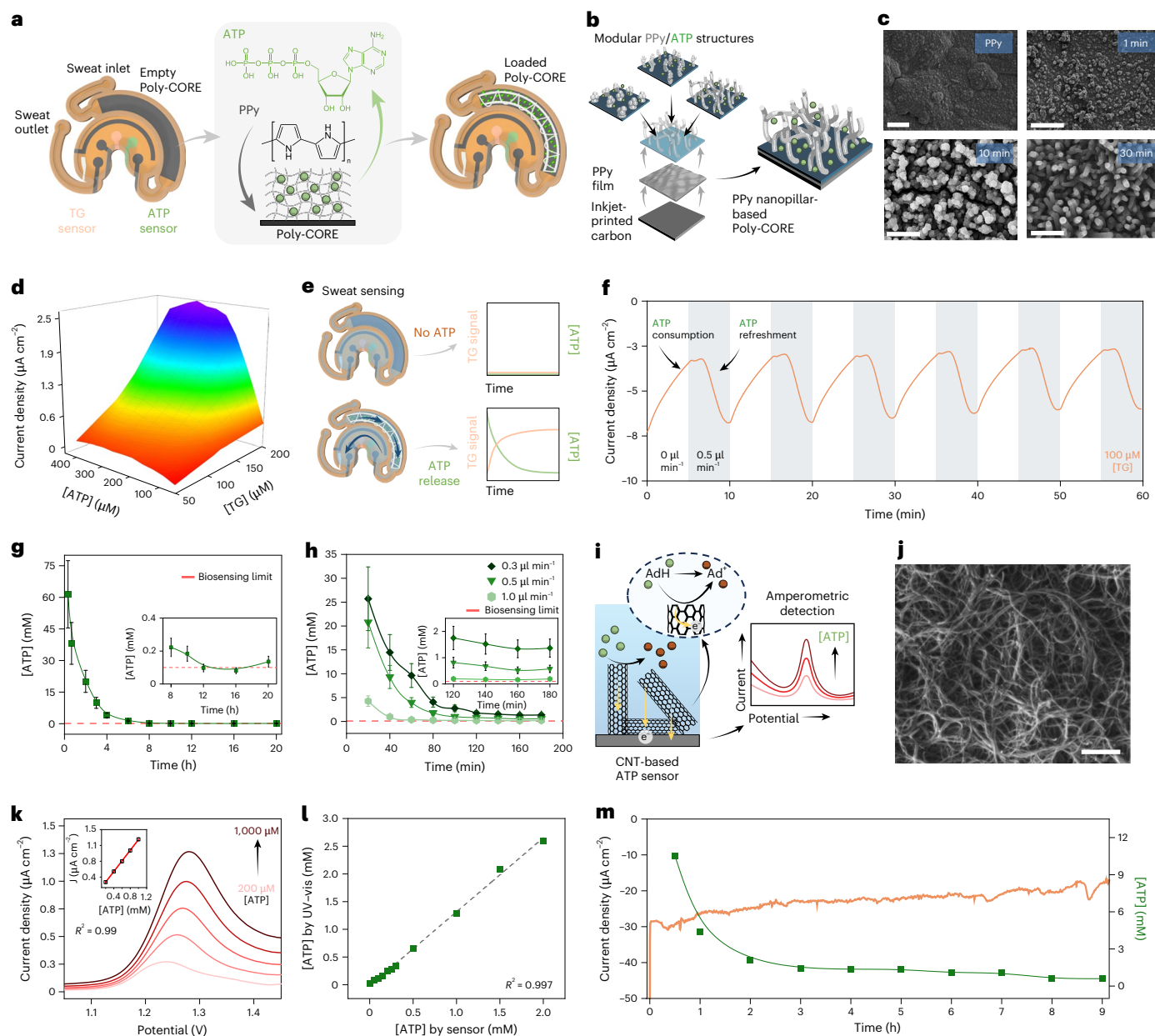


Fig. 3 | Characterization of the Poly-CORE module. **a**, A general schematic illustrating the integration of PPY/ATP nanostructures within the Poly-CORE. **b**, The fabrication workflow showing electrodeposition of PPY/ATP nanopylls on inkjet-printed electrodes. **c**, The SEM images of PPY/ATP nanopylls synthesized for different deposition durations at a constant current density. Scale bar, 30 μm . Representative SEM images are shown from $n = 3$ independently prepared PPY/ATP samples. **d**, A three-dimensional surface plot showcasing the relationship between ATP concentration and the amperometric TG sensor response. **e**, The microfluidic layout of the integrated Poly-CORE module showing sweating-activated ATP diffusion into the TG-sensing chamber. **f**, A dynamic amperometric measurement demonstrating periodic ATP consumption and signal recovery under continuous flow conditions. **g**, The long-term ATP release profile of PPY/

ATP structure (synthesized at 0.15 mA cm^{-2} for 1 h) over 20 h. **h**, The ATP release profile of PPY/ATP nanostructures (synthesized at 0.15 mA cm^{-2} for 10 min) under different flow rates mimicking physiological SRs. Error bars in **g** and **h** represent s.d. from the mean of $n = 3$ independent Poly-CORE modules. **i**, A schematic of the MWCNT-based sensor for direct cofactor quantification. AdH and Ad⁺ represent the reduced and oxidized forms of the adenosine moiety. **j**, A SEM image of MWCNT electrode surface. Scale bar, 400 nm. The representative SEM image is shown from $n = 3$ independently prepared MWCNT electrodes. **k**, The DPV curves for ATP detection (200–1,000 μM). **l**, The validation of ATP measurements against UV-vis at 260 nm. **m**, The representative time course of in vitro TG and ATP detection demonstrating real-time cofactor-refreshing biosensing.

matrix, release can be electrochemically stimulated. Application of a -0.8 V potential for 15 min after 1 h of passive release increased ATP concentrations from $\sim 450 \mu\text{M}$ to $\sim 1 \text{ mM}$ (Supplementary Fig. 28), demonstrating adaptive control of cofactor availability.

To enable real-time calibration, we developed an MWCNT-based ATP sensor that detects ATP oxidation via differential pulse voltammetry (DPV) (Fig. 3i). Incorporating MWCNT onto the inkjet-printed

carbon electrodes enhanced electron transfer (Fig. 3j and Supplementary Fig. 29), whereas a Nafion coating improved analyte preconcentration and sensitivity. The sensor exhibited a clear oxidation peak near 1.25 V and a sensitivity of $1.29 \mu\text{A cm}^{-2} \mu\text{M}^{-1}$ (Fig. 3k and Supplementary Fig. 30) maintaining high accuracy across varying pH values (Supplementary Fig. 31). Selectivity testing in vitro and in human sweat showed minimal interference from common sweat constituents at the

ATP oxidation potential (Supplementary Fig. 32). Validation against UV–visible spectrophotometry (UV–vis) confirmed excellent agreement between methods (Fig. 3l).

Combining the TG enzymatic sensor, Poly-CORE ATP release system and MWCNT ATP sensor enabled continuous TG detection with real-time cofactor calibration (Fig. 3m). Direct ATP measurement provided feedback on cofactor availability and identified conditions where depletion could affect performance. Correlating ATP concentration with TG sensor output enabled an interpolation-based calibration model that compensates for temporal ATP variation (Supplementary Fig. 33, Supplementary Note 3 and Supplementary Table 1). This model was constructed over the operational ATP range produced by the optimized Poly-CORE protocol used for continuous TG sensing, whereas higher ATP concentrations from release-capacity studies were used to define reservoir limits rather than the TG calibration domain. The TG response at elevated ATP concentrations and integrated dynamic sensing at skin temperature were evaluated separately (Supplementary Figs. 34 and 35). This integrated system establishes a generalizable strategy for continuous cofactor-refreshing biosensing in wearable bioelectronics.

Continuous epidermal lipid profiling

The wearable device enables the continuous detection of Glu, CH and TG for comprehensive assessment of cardiometabolic health (Fig. 4a and Supplementary Figs. 36–38). To demonstrate practical on-body performance, we evaluated dynamic epidermal lipid responses under different dietary conditions (Fig. 4b). Sweat was locally induced using a miniaturized IP module, which applies a low-intensity current across two hydrogel electrodes in contact with the skin. The current drives diffusion of the cholinergic agent carbachol into the dermis, activating sweat glands via the cholinergic pathway (Fig. 4c). As described previously⁴¹, this process yields continuous sweat flow that refreshes both the sensor chamber and SR microchannels (Fig. 4d).

Cardiometabolic health is influenced by genetics, metabolic rate, insulin sensitivity and lifestyle, with dietary composition playing a central role in lipid regulation⁴². Nutritional balance directly affects blood sugar and lipid metabolism, impacting both disease prevention and management^{4,43}. To evaluate diet-dependent metabolic dynamics, we continuously profiled biomarker levels during and after the consumption of four meal types, with the fasting state served as a control (Fig. 4e): a lipid-rich meal (Fig. 4f), a mixed macronutrient meal (Fig. 4g, pizza), a protein-dense meal (Fig. 4h) and a high-lipid, high-carbohydrate liquid drink (Fig. 4i, milkshake). SR during the fasting was lower than in postprandial states, probably reflecting reduced hydration status. Paired fasting and postprandial measurements of CH and TG in sweat and blood were performed for a representative condition, showing consistent directional changes between the two matrices (Supplementary Fig. 39). Each meal was consumed ~30 min before sensor placement.

The epidermal sensors captured distinct postprandial trends. It is important to note that the temporal profiles of sweat CH and TG are not governed solely by passive diffusion from blood; similar peak timing relative to Glu does not imply comparable transport kinetics, but rather reflects the combined effects of transport, secretion and partitioning processes within the skin and sweat matrix. Glu levels rose sharply following the mixed macronutrient meal and high-lipid, high-carbohydrate drink, consistent with their higher glycaemic index and rapid absorption of simple sugars. By contrast, meals enriched in fats, CH and protein elicited minimal changes in sweat Glu owing to their lower carbohydrate content.

Both CH and TG levels demonstrated pronounced variability across meal types. During fasting, CH and TG levels gradually decreased, suggesting localized lipid depletion within the sweat glands and epidermis due to slow diffusion from deeper tissue. By contrast, postprandial lipid-rich, mixed, protein-rich and liquid meals induced substantial

elevations in both biomarkers. These dynamic responses probably result from a combination of lipid diffusion from circulation, lipoprotein remodelling and efflux and enhanced sebaceous lipid secretion that mixes with eccrine sweat^{7,44,45}.

The temporal patterns of CH and TG also reflected the meal composition. High-fat meals such as bacon and eggs (Fig. 4f) or ground beef (Fig. 4h) produced sustained increases in CH and TG. The high-fat, high-carbohydrate drink (Fig. 4i) induced a faster, transient response owing to rapid gastrointestinal absorption, with the measurements capturing a later stage of the postprandial stage in which CH and TG levels begin to decline. The pizza meal (Fig. 4g), rich in both carbohydrates and fats, elicited the largest combined rise in epidermal CH and TG (Supplementary Fig. 40), probably driven by hepatic lipoprotein production following digestion.

Individual physiological factors such as age, body composition, fitness and prior meals also contributed to interparticipant variability^{46,47}. Such personalized metabolic signatures highlight the value of continuous monitoring. By capturing real-time fluctuations in lipid and Glu biomarkers, this wearable platform provides individualized insights into diet-induced metabolic responses and enables personalized cardiometabolic health assessment (Fig. 4j).

Lipid biomarker validation via causal ML

Translating non-invasive biomarkers into clinically relevant indicators requires rigorous validation against established standards. We therefore conducted a preclinical study using a causal ML framework to derive a more accurate mapping between sweat and blood lipid concentrations by integrating SR and individual physiological factors (Fig. 5a). Direct sweat–blood comparisons for CH and TG yielded poor correlations (Fig. 5b,c), probably owing to the limited solubility of hydrophobic lipids²⁵, barrier-limited transport across skin and glandular membranes²⁶ and local lipid secretion from sebaceous glands⁴⁸ and interindividual physiological variability. These weak direct correlations highlight the limitations of pairwise relationships and motivate a causal ML framework that jointly models sweat biomarkers and physiological confounders. Accordingly, we collected demographic and physiological data from 24 participants (Fig. 5d) to identify potential candidate covariates for causal adjustment (Supplementary Table 2).

Pearson correlation screening between each collected variable and the corresponding blood lipid endpoint. Using a loose prioritization threshold ($|r| \geq 0.2$), several physiological variables showed non-negligible association with blood lipid levels, whereas direct sweat–blood lipid associations remained comparatively weak (Fig. 5e,f). As this screening analysis evaluates each variable independently, it does not capture how participant-level covariates may modify sweat–blood lipid relationships. For example, although body mass index (BMI) showed only a modest marginal association with blood CH, it remained informative in the causal model by accounting for interindividual differences in the sweat CH–blood CH mapping. These observations motivated a directed acyclic graph (DAG)-based framework to account for physiological variability beyond raw sweat biomarkers.

Task-specific DAGs were constructed by combining the Pearson screening with physiological knowledge of sweat secretion, lipid transport and body composition. SR was treated separately as a dynamic secretion-related factor, whereas static body-composition effects were represented parsimoniously by BMI, with sex additionally retained for the CH task (Fig. 5g). The resulting adjustment sets satisfied the backdoor criterion, ensuring that major confounding pathways were controlled in downstream modelling.

Using these causal structures, we trained models to estimate blood CH and TG levels from sweat data. Model inputs were sweat lipid measurements, SR and participant-level physiological covariates, whereas paired blood lipid measurements were used only as clinical reference labels for model training and validation. Blood lipid measurements

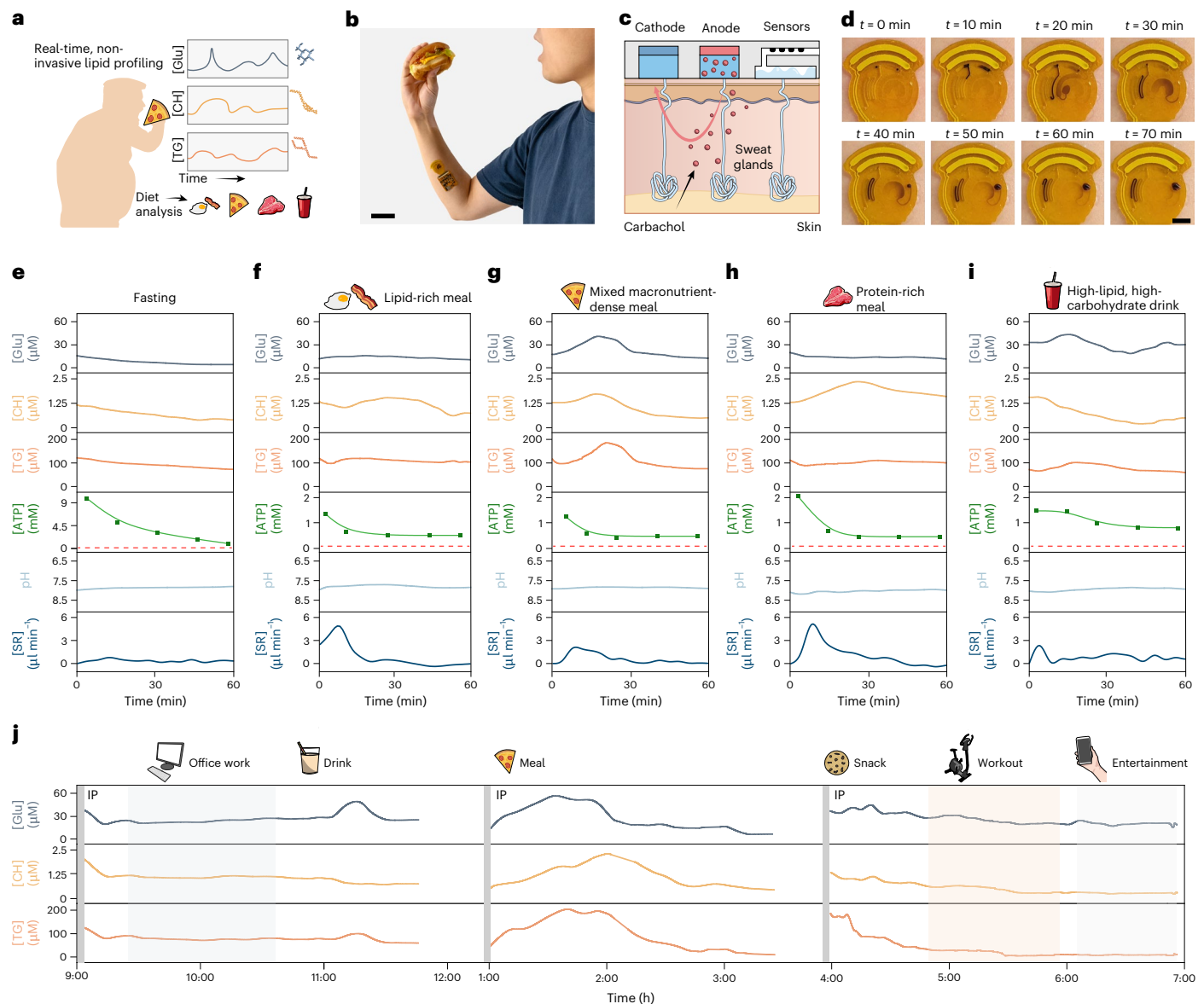


Fig. 4 | Continuous epidermal lipid profiling. **a**, A schematic overview of on-body lipid profiling during different dietary challenges. **b**, An optical image of integrated wearable patch attached to the forearm of a participant. **c**, A conceptual illustration of the IP module within the microfluidic system for localized sweat induction and collection. **d**, Sequential optical images showing sweat filling and refreshment in the designed microfluidic sensor chamber

and SR channels following IP. Scale bar, 3 mm. **e–i**, The continuous epidermal profiling of Glu, CH and TG under different dietary conditions: fasting (**e**), lipid-rich meal (**f**), mixed macronutrient-dense meal (**g**), protein-rich meal (**h**) and high-lipid, high-carbohydrate drink (**i**). **j**, The long-term on-body monitoring of Glu, CH and TG during daily habitual activities.

were not used as prediction inputs. This distinction is important because the goal of the model is to learn sweat-to-blood mapping, rather than to reproduce blood measurements from blood-derived inputs. The causal models achieved strong predictive performance, with R^2 of 0.912 for TG and R^2 of 0.610 for CH, together with low mean absolute errors (MAEs) (16.44 and 15.27 mg dl⁻¹, respectively) (Fig. 5h,i). Compared with baseline linear and regularized regressions, dual-variable causal models incorporating both SR and sweat lipid levels produced consistently lower MAE and higher R^2 values.

To assess whether predictive performance was driven primarily by demographic and physiological covariates, we performed a negative-control analysis using covariates alone while excluding sweat CH and TG measurements (Supplementary Table 3). Removing sweat biochemical inputs markedly reduced performance. For TG, model performance decreased from an MAE of 16.44 and an R^2 of 0.912 in the full model to an MAE of 45.46 and an R^2 of 0.301 in the covariate-only model. For

CH, performance decreased from an MAE of 15.27 and an R^2 of 0.610 to an MAE of 23.49 and an R^2 of 0.062. These results indicate that demographic and physiological covariates alone are insufficient to recover blood lipid levels with comparable accuracy and that inclusion of sweat lipid measurements substantially improves prediction. This distinction is particularly important for TG prediction, where BMI captures static between-participant differences, whereas continuous sweat TG provides time-varying biochemical information. The improvement was more pronounced for CH, where unadjusted correlations were weak owing to lipid transport and partitioning constraints, whereas TG prediction benefited primarily from refinement of moderate associations (Supplementary Table 4).

The causal framework also highlights physiological relationships among the sweat-domain variables. For CH, the weak negative correlation between sweat CH and SR ($r = -0.11$) reflected mild dilution at higher sweating rates. For TG, the inverse correlation ($r = -0.31$)

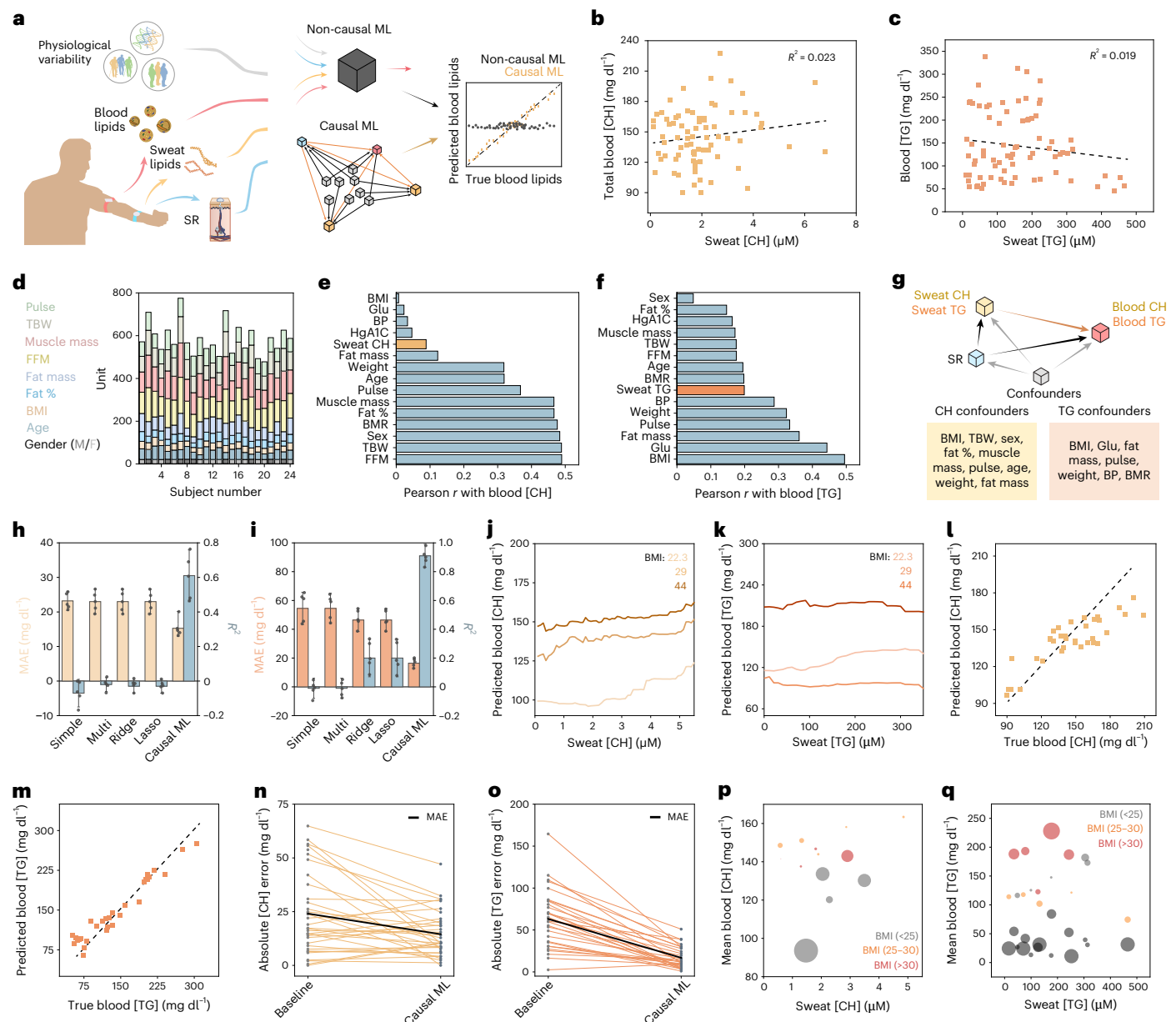


Fig. 5 | Validation of sweat lipid biomarkers using causal ML. a, An overview of the lipid validation workflow integrating sweat and blood analyses with causal ML. **b,c**, Raw bivariate scatter plots showing direct pairwise relationships between sweat and blood CH (**b**) and TG (**c**), with linear-fit R^2 values. **d**, The demographic and physiological characteristics of the 24 participants included in the validation study. Each column represents one participant and is stacked to display multiple variables. The stacked visualization illustrates interparticipant variability across these attributes and does not represent a combined quantitative metric. **e,f**, The Pearson correlation screening results showing marginal Pearson correlation coefficients (r) between each candidate demographic and physiological variable and blood CH (**e**) or TG (**f**), used for covariate prioritization and interpreted separately from the model-based response curves in **j** and **k**. **g**, The task-specific DAGs depicting the hypothesized causal relationships among sweat biomarkers, dynamic SR effects, physiological covariates and blood lipid outcomes. **h,i**, The model performance comparisons for CH (**h**) and TG (**i**) predictions across linear regression, multivariate regression, regularized regressions (ridge and lasso) and causal ML models showing improved MAE and coefficient of determination (R^2) under causal adjustment. Bars represent mean values across $n = 5$ model evaluation runs, overlaid points represent individual runs and error bars represent s.d. Model type was the

comparison group, with linear, multivariate, ridge and lasso regression serving as baseline models for comparison against the causal ML model. **j,k**, Individualized model response curves for CH (**j**) and TG (**k**), showing how predicted blood lipid levels change as sweat lipid inputs vary under fixed participant-level physiological covariates. These curves represent model-predicted responses and do not represent continuously measured blood lipid trajectories. **l,m**, Agreement between model-predicted blood CH (**l**) and TG (**m**) values generated from sweat-based inputs and paired blood lipid measurements used as clinical reference labels. Blood lipid measurements were used for model training and validation, not as prediction inputs. **n,o**, The per-participant change in MAE for CH (**n**) and TG (**o**) showing improved accuracy for most participants under the causal model. Each point represents one participant, and the participant was the independent unit of study. **p,q**, The stratified evaluation of CH (**p**) and TG (**q**) models across BMI categories (normal weight, overweight and obesity), with size indicating the number of participants and colour denoting BMI category. Causal modelling disentangled demographic confounding, revealing BMI-dependent bias in sweat–blood CH relationships and robust, BMI-independent TG coupling, underscoring the superior generalizability of the causal framework across diverse populations.

suggested faster turnover and clearance, supporting the inclusion of SR as an informative physiological covariate.

At the individual level, causal adjustment reduced unexplained interparticipant variability and produced more consistent participant-specific response patterns (Fig. 5j,k). These panels visualize individualized model response curves showing how the trained model predicted blood lipid output changes as sweat lipid input varies under fixed participant-level physiological covariates; they do not represent directly measured continuous blood lipid trajectories. When evaluated against paired blood measurements collected at discrete clinical sampling points, the causal model demonstrated improved agreement between predicted and measured blood lipid concentrations (Fig. 5l,m and Supplementary Fig. 41) and reduced per-participant errors (Fig. 5n,o). These gains persisted across BMI-stratified subgroups, including participants in the normal-weight, overweight and obesity BMI categories, indicating subgroup generalizability (Fig. 5p,q). Bootstrap resampling ($n = 1,000$) and subgroup-specific analyses further supported the stability of the causal estimates while preserving meaningful population-level heterogeneity.

SR consistently emerged as a global marker of metabolic activity and dilution, whereas sweat CH showed stronger dependence on BMI and sex. By contrast, sweat TG correlated more directly with blood TG, supporting the notion that TG excretion reflects systemic lipid clearance process rather than a demographically confounded phenomenon. Although SR was the dominant physiological variable in the present framework, additional biochemical or physiological factors, including pH, may also contribute to interindividual variability; however, pH variation remained limited under carbachol-based IP conditions in this study and our prior reports^{49–51}.

Overall, integrating causal structure and confounder control improved the ability of non-invasive sweat lipid measurements to reflect underlying blood lipid physiology. For CH estimation, adjustment for BMI and sex mitigated systematic bias, with SR serving as a physiological calibrator. For TG, the joint modelling of sweat TG and SR produced robust predictions across BMI groups. These findings support causal ML-guided sweat lipid monitoring as a framework for continuous, personalized metabolic assessment.

Conclusion

This study demonstrates a wearable epidermal lipid profiler for the continuous, non-invasive monitoring of Glu, CH and TG. Biosensing was achieved through single-step, dual-step and cascading enzymatic reactions, respectively, with TG detection uniquely requiring ATP as an external cofactor. To sustain this reaction, we developed and integrated a Poly-CORE module within the microfluidic sensing chamber. Coupled with a MWCNT-based ATP biosensor, the platform enables real-time cofactor calibration and long-term stable operation. The tunable Poly-CORE architecture extends continuous biosensing to enzymatic systems that rely on non-regenerable cofactors.

The platform also incorporates a carbachol-based IP module for autonomous sweat induction, alongside integrated pH and SR sensors for physiological calibration. On-body monitoring revealed dynamic lipid fluctuations in response to dietary inputs and habitual activities. In a 24-participant preclinical study, simultaneous sweat and blood lipid analyses combined with causal ML improved the mapping accuracy between non-invasive and systemic lipid levels.

Although the present dataset included repeated longitudinal sweat–blood measurements, the number of independent participants remained modest. Accordingly, the current causal ML analysis should be interpreted as a preclinical proof-of-concept rather than a definitive population-scale model. Future studies with larger and more diverse cohorts, including simultaneous continuous blood and sweat lipid sampling, will be needed to validate robustness, clarify real-time lipid kinetics and assess the utility of continuous epidermal lipidomics for precision cardiometabolic health monitoring.

Methods

Materials

Polyimide (PI) films (–12–75- μm thick) were purchased from DuPont. PET films (–12–250- μm thick) were purchased from McMaster-Carr. Medical adhesives (M-tapes, 3M 468MP) were purchased from 3M. ChEs (123 U mg^{-1}), ChOx (18.2 U mg^{-1}), LIP (33.3 U mg^{-1}), GK (102 U mg^{-1}) and G3PO (23.8 U mg^{-1}) were purchased from Toyobo. Potassium chloride (KCl), nickel chloride (NiCl_2), potassium(III) ferricyanide ($\text{K}_3[\text{Fe}(\text{CN})_6]$), agarose, carbachol, GOx from *Aspergillus niger* (216 U mg^{-1}), BSA, polyvinyl butyral resin BUTVAR B-98 (PVB), aniline, iron(III) chloride, sodium hydroxide, sodium chloride (NaCl), lithium perchlorate, CR monohydrate and PPy were purchased from Sigma-Aldrich. GT (25% aqueous solution), UA and choline chloride (CHI) were purchased from Alfa Aesar. Amplex Red CH Assay kit, tetrachloroauric acid (HAuCl_4), 10 $\times$ phosphate-buffered solution (PBS), hydrochloric acid (HCl), methanol (MeOH), adenine 5-triphosphate disodium salt hydrate (ATP) and dextrose (D-Glu) anhydrous were purchased from Thermo Fisher Scientific. The TG quantification assay was purchased from Abcam. Triolein was purchased from Tokyo Chemical Industry.

Fabrication and assembly of wearable sensors

Printed electrodes. PI films served as the substrate for inkjet printing (DMP-2850, Fujifilm). Silver ink was utilized to print the interconnects, connection pads, interdigitated SR sensor electrodes and reference electrode. Carbon ink was used to print the IP electrodes, counter electrode and working electrodes (on top of the connection pads). The reference and working electrodes were further modified for electrochemical sensing using an electrochemical station (CHI 840D).

Microfluidics. A 50 W CO_2 laser cutter (Universal Laser System) was used to pattern the microfluidic layers. M-tape (3M 468MP) and PET layers were used for layered patterning on the basis of the specifications from the computer-aided designs. The fabricated microfluidic layers were assembled onto the PI substrate for sweat collection and transport.

IP gels. The IP anode and cathode gels were prepared by heating deionized (DI) water containing 3% w/w agarose at 200 $^{\circ}\text{C}$ under continuous stirring to create homogeneous solutions. The solutions were then cooled to 165 $^{\circ}\text{C}$. Once at 165 $^{\circ}\text{C}$, carbachol (1% w/w) was added to the anode mixture, whereas KCl (1% w/w) was added to the cathode mixture. The prepared mixtures were then poured into the IP electrode moulds of the assembled microfluidic sensor patches.

Biosensor development and characterization

Enzymatic sensors. Gold nanodendrites were grown on the surface of the carbon working electrodes by applying pulsed voltage from –0.9 to 0.9 V for 1,500 cycles at a frequency of 50 Hz using a 50 mM tetrachloroauric acid and 50 mM HCl solution. A PB redox mediator was then electrochemically deposited on the surface of the Au layer by performing cyclic voltammetry from –0.2 to 0.6 V at a scan rate of 50 mV s^{-1} in a solution containing 2.0 mM iron(III) chloride, 2.5 mM $\text{K}_3[\text{Fe}(\text{CN})_6]$, 0.1 M KCl and 0.1 M HCl for 20, 6 and 20 cycles corresponding to the Glu, CH and TG sensors. Last, a NiHCF layer was deposited on the surface of the PB using cyclic voltammetry from 0 to 0.8 V at a scan rate of 100 mV s^{-1} for eight cycles in a solution containing 5.0 mM NiCl_2 , 2.5 mM $\text{K}_3[\text{Fe}(\text{CN})_6]$, 0.1 M KCl and 0.1 M HCl.

The enzyme solution for the Glu sensor was prepared by mixing 99.0 μl of 1.0% BSA, 1.0 μl of 2.5% GT and 2.0 μl of 10 mg ml^{-1} GOx in 1 $\times$ PBS. The CH sensor enzyme solution was prepared by mixing 8.40 μl of 1.0% BSA, 0.50 μl of 1.0% GT, 3.3 μl of 1.62 mg ml^{-1} ChEs and 3.3 μl of 10.98 mg ml^{-1} ChOx in 1 $\times$ PBS. The enzyme solution for the TG sensor was prepared by mixing 17.5 μl of 1.0% BSA, 0.50 μl of 2.5% GT, 5.8 μl of 36 mg ml^{-1} LIP, 5.8 μl of 6 mg ml^{-1} GK and 5.8 μl of 9.6 mg ml^{-1} G3PO in 1 $\times$ PBS. For all enzymatic sensors, 3 μl enzyme cocktail was then

drop-cast on the surface of the modified electrodes and left to dry overnight at 4 °C.

pH sensor. A gold layer was deposited on the surface of a carbon working electrode by applying a constant potential of 0 V for 30 s while submerged in a 50 mM tetrachloroauric acid and 50 mM HCl solution. After, a polyaniline layer was electropolymerized on the surface of the modified electrode via cyclic voltammetry sweeps from −0.2 to 1.0 V at a scan rate of 50 mV s^{−1} for 50 cycles.

ATP sensor. A 0.01% w/w carbon nanotube (CNT) solution in ethanol was prepared via probe sonication for 2 h to form a uniform dispersion of CNTs. Then, 2 µl of the CNT solution was drop-cast onto an inkjet-printed carbon working electrode. Last, 1 µl of a Nafion solution (0.1% v/v in ethanol) was drop-cast onto the sensor electrode surface and dried overnight.

Reference electrode. First, 0.3 µl of 0.1 M FeCl₃ solution was drop-cast on the surface of the inkjet-printed Ag reference electrode for 30 s to form a Ag/AgCl layer and then rinsed thoroughly with DI water. Then, 1.5 µl of a reference solution, made by sonicating 79.1 mg PVB and 50.0 mg NaCl in 1 ml methanol for 2 h, was drop-cast on the surface of the Ag/AgCl electrode. The modified reference electrode was left to dry overnight.

In vitro biosensor characterization. The in vitro characterization of the biosensors was performed using an electrochemical workstation (CHI840E, CH Instruments for single electrode experiments and a CHI 1430, CH Instruments for multichannel studies) at room temperature. A commercial platinum wire was used as the counter electrode and a commercial Ag/AgCl electrode was used as the reference electrode. Chronoamperometric studies to characterize the enzymatic biosensors were performed at a constant potential of 0 V.

For multichannel experiments, amperometric sensors (Glu, CH and TG) were operated continuously at 0 V versus Ag/AgCl, whereas potentiometric sensors were recorded under open-circuit conditions to prevent current injection into the electrolyte. ATP measurements were performed intermittently using DPV with brief scans (approximately 20–30 s) every 10 min, enabling time-multiplexed operation. The DPV were used under the following conditions: potential range, 0.8–1.6 V; incremental potential, 0.04 V; pulse amplitude, 0.05 V; pulse width, 0.05 s; and pulse period, 0.5 s. The counter electrode was designed to maintain uniform spacing relative to each working electrode, thereby minimizing unintended current pathways and reducing electrochemical coupling between sensing channels. In addition, because analyte concentrations are in the low micromolar range, the corresponding H₂O₂ production at each sensing site is inherently low, further limiting its potential to perturb neighbouring electrodes or induce measurable pH shifts, with any generated H₂O₂ further reduced at the PB transducer interface.

The Glu, CH and TG sensors were conditioned in PBS until a stable signal was observed, and then, the solutions were spiked to achieve concentrations of 0–200 µM of Glu, 0–3 µM of CH and 0–150 µM of TG, respectively. For the TG sensor testing in vitro, the buffer solution contained 200 µM of ATP, to realize the enzymatic reaction. The pH dependency studies were performed by altering the solution's pH (from 7 to 9). Selectivity studies were performed by spiking the sensors with a known concentration of their corresponding target (100 µM, 3 µM and 50 µM for the Glu, CH and TG sensors, respectively) and spiking known concentrations of the potential interferents within the physiological ranges found in sweat. Shelf stability was accessed by storing an array of biosensors at 4 °C and testing a new subset ($n = 4$) at weekly intervals. The characterization of the pH sensor was achieved by performing open-circuit potential studies using McIlvaine buffers with pH values ranging from 3 to 8. The pH of each buffer solution was independently

measured using a calibrated commercial pH metre before testing. The open-circuit potential of the integrated pH sensor was recorded after stabilization at each pH value and calibration curves were constructed to evaluate sensitivity and linearity.

Sweating rate sensor and characterization. For in vitro flow tests, a syringe pump (Braintree Scientific) was used to flow various solutions into the microfluidic channels at different flow rates (0.25–3 µl min^{−1}). NaCl solutions with various concentrations (12.5–200 mM) were prepared in DI water. For the characterization of the SR sensor under various pH, McIlvaine buffers were used. A ceramic hot plate (Thermo Fisher Scientific) was used to characterize dependence on temperature in SR sensor.

Poly-CORE module synthesis and characterization

The PPy/ATP structures within the CORE unit were synthesized using an electrochemical workstation (CHI840E, CHI Instruments) at room temperature. A commercial platinum wire was used as the counter electrode, and a commercial Ag/AgCl electrode was used as the reference electrode. First, an electrolyte solution containing 0.14 M pyrrole and 0.07 M lithium perchlorate was prepared in DI water. This solution was utilized to deposit a thin PPy film onto the surface of an inkjet carbon electrode by applying a constant voltage of 1.0 V for 150 s. To achieve the PPy/ATP structures, different current densities were ramped up (from 0.075 to 1.0 mA cm^{−2}) in a solution containing 0.07 M lithium perchlorate, 0.20 M ATP and 0.15 M pyrrole, until a voltage of 0.6 V or higher was achieved to ensure polymerization. The total electrodeposition time for the characterization and on-body experiments was 30 min.

SEM images were taken to observe the morphological differences of the PPy/ATP structures synthesized in different electrodeposition conditions. AC impedance studies were performed in a 0.1 NaCl solution using 0.01 V amplitude at a frequency range of 100 kHz to 1 Hz in a CHI 660 electrochemical station. A Braintree Scientific pump was utilized for all the in vitro ATP-releasing and TG-sensing experiments. For the ATP release profile characterizations of the PPy/ATP structures, 1× PBS samples were collected from the microfluidic outlet of the sensing chamber to quantify the ATP concentrations of the samples at different time intervals. The aforementioned samples were measured via UV–vis (Thermo Scientific NanoDrop) at a wavelength of 260 nm. The PPy/ATP structures used for on-body experiments were electrodeposited the night before testing to ensure fresh ATP loading. After fabrication, the devices were stored at 4 °C and protected from light until use. All ATP-containing precursor solutions were freshly prepared before each synthesis.

On-body evaluations of wearable platform and validation

Participant recruitment. The validation and evaluation of the sweat sensor were performed using human participants in compliance with all the ethical regulations under a protocol (ID 19-0892) that was approved by the Institutional Review Board at the California Institute of Technology. Participants were recruited from the campus and neighbouring California Institute of Technology and the University of California at Los Angeles communities. Participants from ages 21 to 65 years were recruited. Normal weight individuals with a BMI of 18.5–24.9 kg m^{−2} with fasting serum Glu <100 mg dl^{−1} were enrolled. Individuals with overweight or obesity, defined as a BMI of 25–35 kg m^{−2}, were also eligible if they had fasting serum Glu ≥100 mg dl^{−1} and <126 mg dl^{−1} or HbA1c >5.7% and <6.5% or if they had fasting serum Glu ≤100 mg dl^{−1} or HbA1c <5.7%.

Participants with a screening laboratory value outside of the laboratory normal range that was considered clinically notable for study participation by the investigator, taking blood thinning medications such as warfarin or coumadin, using tobacco products, unable or unwilling to comply with the study protocol, with a history of diabetes

mellitus or other serious medical conditions as indicated by medical history and participants that were pregnant or breastfeeding were excluded from the study.

All participants gave written informed consent before participating in the study. Participant age, sex and other demographic and physiological characteristics for the validation cohort are provided in Supplementary Table 2. Participants were eligible to receive up to US\$150 for completion of the study, including US\$25 for screening and US\$125 for the test day.

On-body evaluations. On-body evaluations were performed using a multichannel electrochemical workstation (CHI 1430, CH Instruments). For the fasting study, the participant did not consume a meal for 12 h before the beginning of the experiment. All the meals were consumed 30 min before placing the wearable device. The forearm of the participant was cleaned with an ethanol swab and DI water before attaching the sensing patch. IP was run for 5 min at a constant current of 1 mA to achieve localized sweat induction. Once the sweat filled the microfluidic sensing chamber, the measurements started. For independent long-term on-body evaluations, a new sensing patch was used for each session and positioned at a neighbouring site on the forearm to enable reproducible, extended duration sweat collection following IP.

Validation studies. The biomarker validation studies were performed by collecting sweat samples after induction via a Model 3700 Macroduct Sweat Collection System. First, the forearm of the participant was cleaned with an ethanol swab before placing the commercial IP electrodes. After sweat induction, the participant's forearm was subsequently cleaned before placing the sweat collector. Sweat samples were collected for a maximum of 120 min. The collected samples were then analysed using the commercial CH and TG quantification kits.

In parallel to the sweat collection, at the beginning of the study, a peripheral venous catheter was inserted into a suitable vein of each participant. Blood samples were collected through the catheter at 30-min intervals over the course of the study (180 min). These repeated blood draws provided discrete paired reference measurements for model training and validation but do not constitute a continuous temporal gold standard at the same resolution as the sweat sensor. The blood samples were immediately stored for further processing. Participants consumed 75 g of Glu during the study.

Causal inference framework for sweat-based biomarker prediction

Causal DAG construction. Task-specific causal DAGs were developed to represent hypothesized relationships among sweat biomarkers, physiological covariates and blood lipid outcomes (Fig. 5e–g). Candidate variables were assembled from the demographic and physiological measurements collected in the preclinical cohort, including age, sex, BMI, Glu, blood pressure, HbA1c, body composition metrics (fat mass, muscle mass and fat-free mass (FFM)), basal metabolic rate (BMR), total body water (TBW), together with sweat lipid measurements and SR.

Pearson correlation analysis was performed between each candidate variable and the corresponding blood lipid endpoint (blood CH or blood TG) as an exploratory prioritization step. Variables with $|r| \geq 0.2$ were considered for further evaluation, although this threshold was not used as a strict selection criterion.

Final inclusion and placement of variables in the DAGs were determined by combining these empirical trends with physiological domain knowledge, while accounting for multicollinearity and model parsimony. SR was treated separately as a dynamic secretion-related factor, whereas static body-composition effects were represented parsimoniously by BMI. Sex was additionally retained for the CH task on the basis of both empirical trends and known physiological differences in lipid regulation.

The resulting adjustment sets satisfied the backdoor criterion, ensuring that major confounding pathways between treatment and outcome were controlled in downstream modelling.

Backdoor adjustment and model training. Backdoor adjustment followed Pearl's causal hierarchy to estimate unbiased causal effects. Two model architectures were used for each task: (1) dual-variable causal models that incorporated both SR and the corresponding sweat lipid and (2) baseline regressors including simple linear, multivariate linear, ridge and lasso models (Fig. 5h,i). Stratified analyses by BMI group and participant-wise response curves were performed to evaluate heterogeneity and individual-level robustness (Fig. 5j–p). Given the modest cohort size, model dimensionality was intentionally constrained through parsimonious covariate selection. Blood samples were collected at 30-min intervals over 180 min, yielding repeated paired sweat–blood measurements within each participant, which increased the information available for model fitting without being treated as independent observations.

Dual-variable causal modelling. To capture joint effects between secretion dynamics and lipid concentration, two treatment variables were defined: sweat lipid (treatment 1) and sweat secretion rate (treatment 2). The joint causal model estimated their combined influence on blood lipid concentrations, accounting for correlations between the two predictors. This approach allowed the simultaneous evaluation of dilution effects and metabolic turnover alongside concentration effects.

Heterogeneous treatment effect estimation. Subgroup-specific causal effects were calculated using sample-size-weighted correlations within strata defined by demographic or physiological variables. Bootstrap resampling ($n = 1,000$) was used to assess effect stability and to construct confidence intervals. Effect magnitudes were classified as weak ($|r| < 0.1$), moderate ($0.1 \leq |r| < 0.3$) or strong ($|r| \geq 0.3$). Stratified causal estimates were compared with pooled models to assess effect heterogeneity across BMI.

Statistical analysis. Statistical testing was conducted using two-sided t -tests within each stratum, with a significance threshold of $\alpha = 0.05$. Interaction terms such as biomarker $\times$ BMI and biomarker $\times$ sex were included to the test for moderation effects. Model performance was evaluated using MAE and the coefficient of determination (R^2). All analyses were implemented in Python (v3.10) using the scikit-learn and statsmodels packages. Reproducible code pipelines were used, and results were validated through fivefold cross-validation.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request.

Code availability

Custom code used for the causal ML analysis is publicly available via GitHub at <https://github.com/SIJIEJI/polycore-lipid-causal> and is cited in the reference list.

References

1. Huang, Y. et al. Lipid profiling identifies modifiable signatures of cardiometabolic risk in children and adolescents with obesity. *Nat. Med.* **31**, 294–305 (2025).
2. Libby, P. et al. Atherosclerosis. *Nat. Rev. Dis. Primer* **5**, 56 (2019).

3. Powell-Wiley, T. M. et al. Obesity and cardiovascular disease: a scientific statement from the American Heart Association. *Circulation* **143**, e984–e1010 (2021).
4. O'Connor, E. A., Evans, C. V., Rushkin, M. C., Redmond, N. & Lin, J. S. Behavioral counseling to promote a healthy diet and physical activity for cardiovascular disease prevention in adults with cardiovascular risk factors: updated evidence report and systematic review for the US Preventive Services Task Force. *JAMA* **324**, 2076 (2020).
5. Shen, X. et al. Multi-omics microsampling for the profiling of lifestyle-associated changes in health. *Nat. Biomed. Eng.* **8**, 11–29 (2023).
6. Pourafshar, S., Parikh, M., Abdallah, B., Al Thubian, N. & Jacobson, J. W. An assessment of individual preference for a novel capillary blood collection system. *Patient Prefer. Adherence* **18**, 531–541 (2024).
7. Baker, L. B. Physiology of sweat gland function: the roles of sweating and sweat composition in human health. *Temperature* **6**, 211–259 (2019).
8. Brunmair, J. et al. Finger sweat analysis enables short interval metabolic biomonitoring in humans. *Nat. Commun.* **12**, 5993 (2021).
9. Zhao, C. et al. Skin-like drift-free biosensors with stretchable diode-connected organic field-effect transistors. *Nat. Electron.* <https://doi.org/10.1038/s41928-025-01465-4> (2025).
10. Clark, K. M. & Ray, T. R. Recent advances in skin-interfaced wearable sweat sensors: opportunities for equitable personalized medicine and global health diagnostics. *ACS Sens.* **8**, 3606–3622 (2023).
11. Davis, N., Heikenfeld, J., Milla, C. & Javey, A. The challenges and promise of sweat sensing. *Nat. Biotechnol.* **42**, 860–871 (2024).
12. Min, J. et al. Skin-interfaced wearable sweat sensors for precision medicine. *Chem. Rev.* **123**, 5049–5138 (2023).
13. Yang, D. S., Ghaffari, R. & Rogers, J. A. Sweat as a diagnostic biofluid. *Science* **379**, 760–761 (2023).
14. Childs, A. et al. Diving into sweat: advances, challenges, and future directions in wearable sweat sensing. *ACS Nano* **18**, 24605–24616 (2024).
15. Zhong, B. et al. Interindividual- and blood-correlated sweat phenylalanine multimodal analytical biochips for tracking exercise metabolism. *Nat. Commun.* **15**, 624 (2024).
16. Kwon, K. et al. An on-skin platform for wireless monitoring of flow rate, cumulative loss and temperature of sweat in real time. *Nat. Electron.* **4**, 302–312 (2021).
17. Liu, W., Cheng, H. & Wang, X. Skin-interfaced colorimetric microfluidic devices for on-demand sweat analysis. *npj Flex. Electron.* **7**, 43 (2023).
18. Sempionatto, J. R. et al. An epidermal patch for the simultaneous monitoring of haemodynamic and metabolic biomarkers. *Nat. Biomed. Eng.* **5**, 737–748 (2021).
19. Yin, L. et al. A stretchable epidermal sweat sensing platform with an integrated printed battery and electrochromic display. *Nat. Electron.* **5**, 694–705 (2022).
20. Lin, S. et al. Non-invasive touch-based lithium monitoring using an organohydrogel-based sensing interface. *Adv. Mater. Technol.* **8**, 2202141 (2023).
21. Narwal, V. et al. Cholesterol biosensors: a review. *Steroids* **143**, 6–17 (2019).
22. Hooda, V., Gahlaut, A., Gothwal, A. & Hooda, V. Recent trends and perspectives in enzyme based biosensor development for the screening of triglycerides: a comprehensive review. *Artif. Cells Nanomed. Biotechnol.* **46**, 626–635 (2018).
23. Kvasnička, A. et al. SLIDE—novel approach to apocrine sweat sampling for lipid profiling in healthy individuals. *Int. J. Mol. Sci.* **22**, 8054 (2021).
24. Sempionatto, J. R., Lasalde-Ramírez, J. A., Mahato, K., Wang, J. & Gao, W. Wearable chemical sensors for biomarker discovery in the omics era. *Nat. Rev. Chem.* **6**, 899–915 (2022).
25. Smith, K. R. & Thiboutot, D. M. Thematic review series: skin lipids. Sebaceous gland lipids: friend or foe? *J. Lipid Res.* **49**, 271–281 (2008).
26. Vietri Rudan, M. & Watt, F. M. Mammalian epidermis: a compendium of lipid functionality. *Front. Physiol.* **12**, 804824 (2022).
27. Prosperi, M. et al. Causal inference and counterfactual prediction in machine learning for actionable healthcare. *Nat. Mach. Intell.* **2**, 369–375 (2020).
28. Feuerriegel, S. et al. Causal machine learning for predicting treatment outcomes. *Nat. Med.* **30**, 958–968 (2024).
29. Expert Panel On Detection, Evaluation, And Treatment Of High Blood Cholesterol In Adults. Executive summary of the third report of the National Cholesterol Education Program (NCEP) expert panel on detection, evaluation, and treatment of high blood cholesterol in adults (adult treatment panel III). *JAMA J. Am. Med. Assoc.* **285**, 2486–2497 (2001).
30. Miller, M. et al. Triglycerides and cardiovascular disease: a scientific statement from the American Heart Association. *Circulation* **123**, 2292–2333 (2011).
31. Sattar, N., McGuire, D. K. & Gill, J. M. R. High circulating triglycerides are most commonly a marker of ectopic fat accumulation: connecting the clues to advance lifestyle interventions. *Circulation* **146**, 77–79 (2022).
32. Parhofer, K. G. Interaction between glucose and lipid metabolism: more than diabetic dyslipidemia. *Diabetes Metab. J.* **39**, 353–362 (2015).
33. Sabaté Del Río, J., Henry, O. Y. F., Jolly, P. & Ingber, D. E. An antifouling coating that enables affinity-based electrochemical biosensing in complex biological fluids. *Nat. Nanotechnol.* **14**, 1143–1149 (2019).
34. Weber, C. J., Strom, N. E. & Simoska, O. Electrochemical deposition of gold nanoparticles on carbon ultramicroelectrode arrays. *Nanoscale* **16**, 16204–16217 (2024).
35. Matos-Peralta, Y. & Antuch, M. Review—Prussian blue and its analogs as appealing materials for electrochemical sensing and biosensing. *J. Electrochem. Soc.* **167**, 037510 (2020).
36. Ying, S. et al. Synthesis and applications of Prussian blue and its analogues as electrochemical sensors. *ChemPlusChem* **86**, 1608–1622 (2021).
37. Xu, C. et al. A physicochemical-sensing electronic skin for stress response monitoring. *Nat. Electron.* **7**, 168–179 (2024).
38. Song, Y. et al. 3D-printed epifluidic electronic skin for machine learning-powered multimodal health surveillance. *Sci. Adv.* **9**, eadi6492 (2023).
39. Davis, N. et al. Reusable, fully integrated sweat monitor band with peel-and-stick-replacement printed microfluidic sensor. *Adv. Mater. Technol.* **10**, 2500477 (2025).
40. Ru, X. et al. Synthesis of polypyrrole nanowire network with high adenosine triphosphate release efficiency. *Electrochim. Acta* **56**, 9887–9892 (2011).
41. Simmers, P., Li, S. K., Kasting, G. & Heikenfeld, J. Prolonged and localized sweat stimulation by iontophoretic delivery of the slowly-metabolized cholinergic agent carbachol. *J. Dermatol. Sci.* **89**, 40–51 (2018).
42. Wang, P., Gao, X., Willett, W. C. & Giovannucci, E. L. Socioeconomic status, diet, and behavioral factors and cardiometabolic diseases and mortality. *JAMA Netw. Open* **7**, e2451837 (2024).
43. Wu, J. H. Y., Micha, R. & Mozaffarian, D. Dietary fats and cardiometabolic disease: mechanisms and effects on risk factors and outcomes. *Nat. Rev. Cardiol.* **16**, 581–601 (2019).

44. Gugliucci, A. The chylomicron saga: time to focus on postprandial metabolism. *Front. Endocrinol.* **14**, 1322869 (2023).
45. Mucinski, J. M. et al. High-throughput LC–MS method to investigate postprandial lipemia: considerations for future precision nutrition research. *Am. J. Physiol. Endocrinol. Metab.* **320**, 702–715 (2021).
46. Berry, S. E. et al. Human postprandial responses to food and potential for precision nutrition. *Nat. Med.* **26**, 964–973 (2020).
47. Mendes-Soares, H. et al. Assessment of a personalized approach to predicting postprandial glycemic responses to food among individuals without diabetes. *JAMA Netw. Open* **2**, e188102 (2019).
48. Picardo, M., Ottaviani, M., Camera, E. & Mastrofrancesco, A. Sebaceous gland lipids. *Dermatoendocrinol.* **1**, 68–71 (2009).
49. Shin, S. et al. A bioinspired microfluidic wearable sensor for multiday sweat sampling, transport, and metabolic analysis. *Sci. Adv.* **11**, eadw9024 (2025).
50. Tu, J. et al. Stressomic: a wearable microfluidic biosensor for dynamic profiling of multiple stress hormones in sweat. *Sci. Adv.* **11**, eadx6491 (2025).
51. Min, J. et al. An autonomous wearable biosensor powered by a perovskite solar cell. *Nat. Electron.* **6**, 630–641 (2023).

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Author contributions

W.G., J.A.L.-R. and J.R.S. conceived of the project. J.A.L.-R., J.R.S., C.W. and S.J. led the main study and collected the overall data. C.W. and D.M. contributed to data processing. R.M.M. contributed to sensor development. S.S., K.K., G.K., H.H., R.M.M. and Y.S. contributed to sensor characterization and testing. S.S., B.A. and J.Y. contributed to the human studies. T.K.H., Z.L., T.L., J.R.S. and W.G. supervised the studies. W.G., J.A.L.-R., J.R.S., C.W. and S.J. cowrote the paper. All authors contributed to the data analysis and provided feedback on the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Juliane R. Sempionatto or Wei Gao.

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	CH instruments were used for all sensor data collection. Excel was used to collect UV-Vis data from characterizations. A BioTek plate reader was used to collect assay data for sensor validation.
Data analysis	OriginLab 2021 was used to plot the data and calculate statistical parameters. MATLAB 2024 was used for calibration of TG sensor data. Python was used for causal ML analyses.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The data supporting the results of this work are available from the corresponding author upon request.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Self-reported information on the subjects' sex was collected. Sex was one parameter analyzed as a potential confounder of sweat-blood lipid correlations.
Reporting on race, ethnicity, or other socially relevant groupings	Race, ethnicity, or other socially relevant groupings were not relevant in this study.
Population characteristics	Subjects from ages 21 to 65 were recruited. Normal weight individuals with a BMI of 18.5 to 24.9 kg m ⁻² with fasting serum glucose <100 mg dl ⁻¹ were enrolled. Overweight/obese individuals with a BMI of 25 to 35 kg m ⁻² and fasting serum glucose ≥100 mg dl ⁻¹ and <126 mg dl ⁻¹ or HbA1c >5.7 % and <6.5 %; Overweight/obese individuals without prediabetes: BMI of 25 to 35 kg m ⁻² and fasting serum glucose ≤100 mg dl ⁻¹ or HbA1c <5.7 % were also eligible.
Recruitment	Subjects were recruited from the campus and neighboring California Institute of Technology and the University of California at Los Angeles communities.
Ethics oversight	Institutional Review Board (IRB) at California Institute of Technology

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	For wearable sensor evaluation, 1 subject was recruited to participate in the continuous sensing studies as well as the diet effect studies. For clinical validation of the epidermal lipids, 24 subjects were recruited.
Data exclusions	In the clinical validation, data from 1 subject was excluded because the subject produced no sweat for analysis.
Replication	All attempts at replication of the studies were successful when following the fabrication process described in the paper. Characterization experiments were performed in replicates as described in the figure legends and methods section.
Randomization	Randomization is not relevant in this work. All devices were tested were fabricated under the same procedure. All subjects studies were done under the same conditions.
Blinding	Not relevant, a blinding process would not influence the sampling result.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.